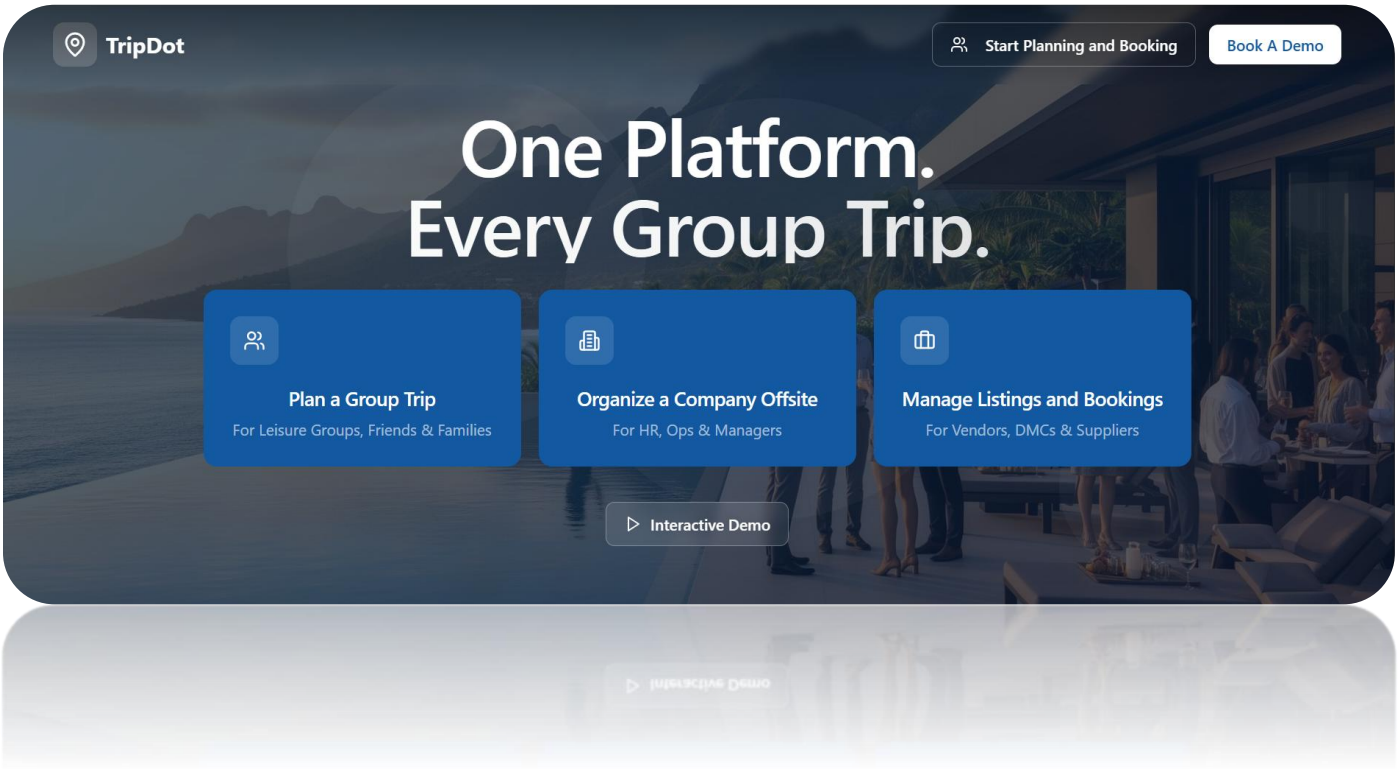
TripDot: Group Travel Planning and Booking Engine



1. Problem Definition

Who: Leisure groups, corporate HR/Ops teams, travelers, and vendors

Struggle with: No unified platform for planning, aligning preferences, approving budgets, booking, and paying for group travel (across leisure trips, corporate offsites, and vendor coordination)

Impact: Wasted time, failed coordination, budget overruns, low vendor conversion, and financial stress for all parties

Success = ≥60% reduction in group trip planning time | ≥80% of corporate offsites completed within budget | Vendor RFP-to-booking rate ≥40%

Stakeholder	Core Pain	Impact
Leisure Organizers (e.g. Group of students)	Planning fragmented across WhatsApp, polls, spreadsheets	Burnout, payment chaos, abandoned trips
Leisure Travelers	No way to split costs, vote, or track live itinerary	Disputes, disengagement, confusion
Corporate HR/Ops (E.g. Offsites)	No workflow: planning → approvals → vendors → invoice	Budget overruns, compliance risk, manual work
Vendors (Hotels, Venues)	Receive vague emails instead of structured RFPs	Low conversion, high overhead, lost revenue

2. Current Approach & Limits

Leisure: WhatsApp → Doodle → OTAs (Expedia, Booking.com) → Venmo/PayPal | 1–4 weeks, often abandoned

Corporate: Notion + Slack + Google Sheets + email for vendor RFPs + TravelPerk/Navan for individual expense | No unified end-to-end workflow

Vendors: Email inbox for vague inquiries → manual quote building → phone follow-ups | No structured demand channel

Limits: No single source of truth; one person absorbs financial risk; corporate tools don't support offsite-specific approval flows; vendors receive low-quality leads

Failure Mode	What Goes Wrong & Consequence
Preference fragmentation	Dates & budgets scattered across tools with no aggregation → trips stall or are abandoned
No group booking	OTAs built for solo checkout → one person absorbs all financial risk upfront
Manual reimbursement	Post-trip Venmo/bank requests with no enforcement → slow, damages relationships
No approval workflow	Corporate teams lack structured budget sign-off mechanism → compliance risk
Unstructured vendor RFPs	Hotels & venues receive vague emails → 10–15% conversion vs. 40%+ potential

3. How AI Helps

AI will: automate the 'messy middle' across all three segments (Leisure group, corporate, vendors): Preference collection, RFP drafting, quote normalization, consensus, approvals, itinerary building, and arrival coordination
Using: preference forms, RFP data, vendor quotes, budget constraints, booking confirmations
Output: consensus summaries, structured RFPs, normalized quote tables, approval briefs, real-time itineraries, split payment requests
Humans decide: final vendor selection, budget approvals, booking confirmation, dispute resolution

Agent	Segment	Input	Output	Human Loop
Group Consensus	B2C	Votes, dates, budgets per traveler	Top destination+date combos with match scores	Final group vote
Itinerary Builder	B2C + B2B	Confirmed bookings, times, addresses	Shared live itinerary for all travelers	Custom additions
RFP Drafting	B2B	Offsite requirements, headcount, budget	Structured RFP sent to vendors	Final review
Quote Normalization	B2B	Raw vendor quotes (email/PDF)	Standardized comparison table	Vendor shortlist
Pref. Aggregation	B2B	Employee travel preference forms	Aggregated report for organizer	Policy exceptions
Approval Summary	B2B	Budget breakdown, vendor quotes	One-page approval brief for manager	Budget sign-off
Vendor Response	Vendor	Inbound RFP details + inventory	Draft quote proposal for vendor	Quote accuracy
Arrival Coordination	B2C + B2B	Flight data, schedule, hotel address	Arrival logistics plan with comms	Disruptions

Quality Testing & Responsible AI

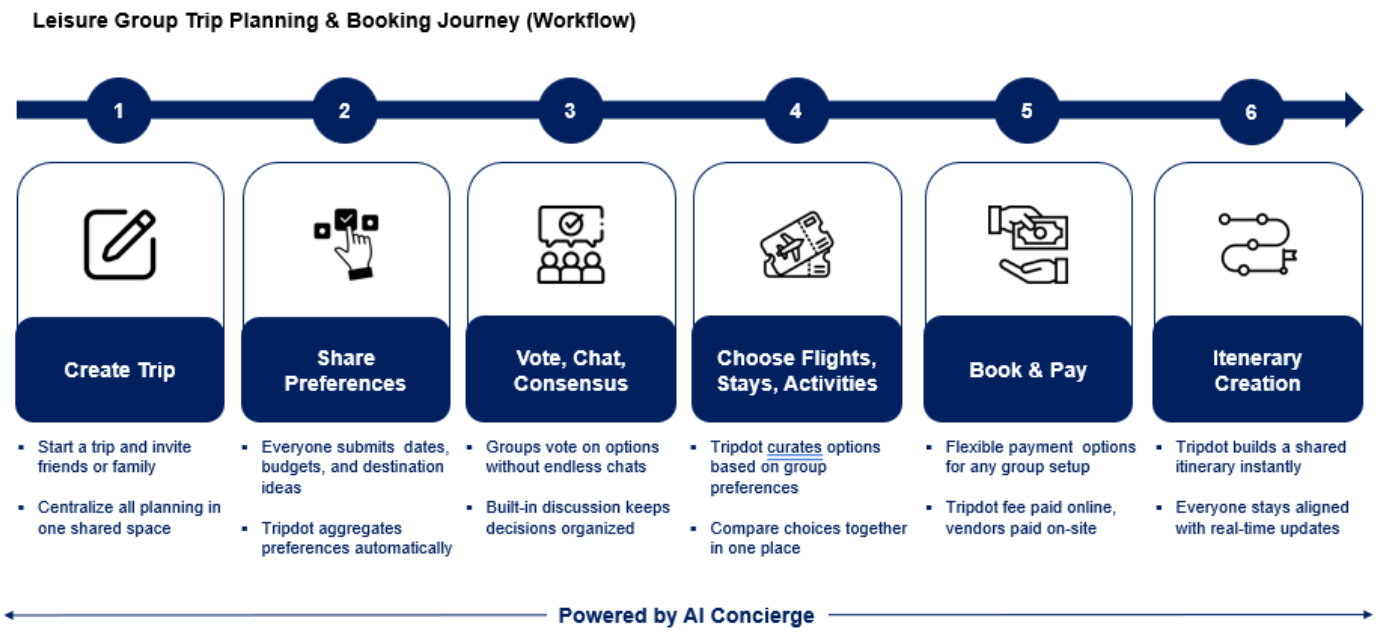
Test Scenario	Method & Threshold (Target)
Preference accuracy	≥95% of key constraints correctly captured (test: 5 simulated forms vs. manual review)
RFP completeness	≥90% field completeness; vendor response rate ≥40% (test: AI vs. human-drafted RFP)
Quote normalization	Zero critical data errors; <5 min vs. ~45 min manual (test: 3 raw vendor quotes)
Itinerary completeness	Zero missing bookings; all times & locations correct in end-to-end test booking
Time-to-booking (B2C)	≥50% reduction vs. WhatsApp + OTA baseline across 5 test groups

Risk	Mitigation
Private budget data	Individual budgets collected privately; never shared between users; only aggregated range visible
Biased AI recommendations	Criteria-based ranking (fit, price, availability) — not data-volume weighted; quarterly audit
Hallucinated vendor quotes	Low-confidence normalizations flagged for human review; structured vendor templates required
Approval bypass risk	All approvals require explicit human action; AI summaries clearly labeled; confidence indicators shown

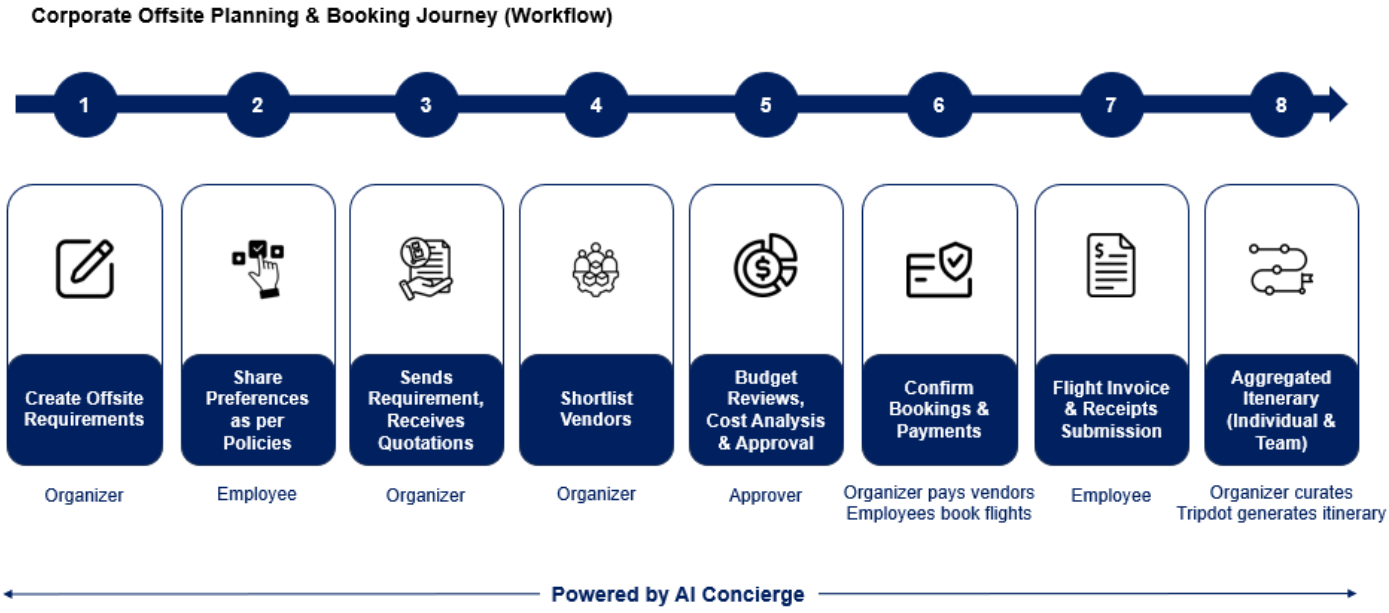
Employee data privacy	Role-based access; employee preferences not shared with vendors; SOC 2 compliance roadmap

4. Workflows

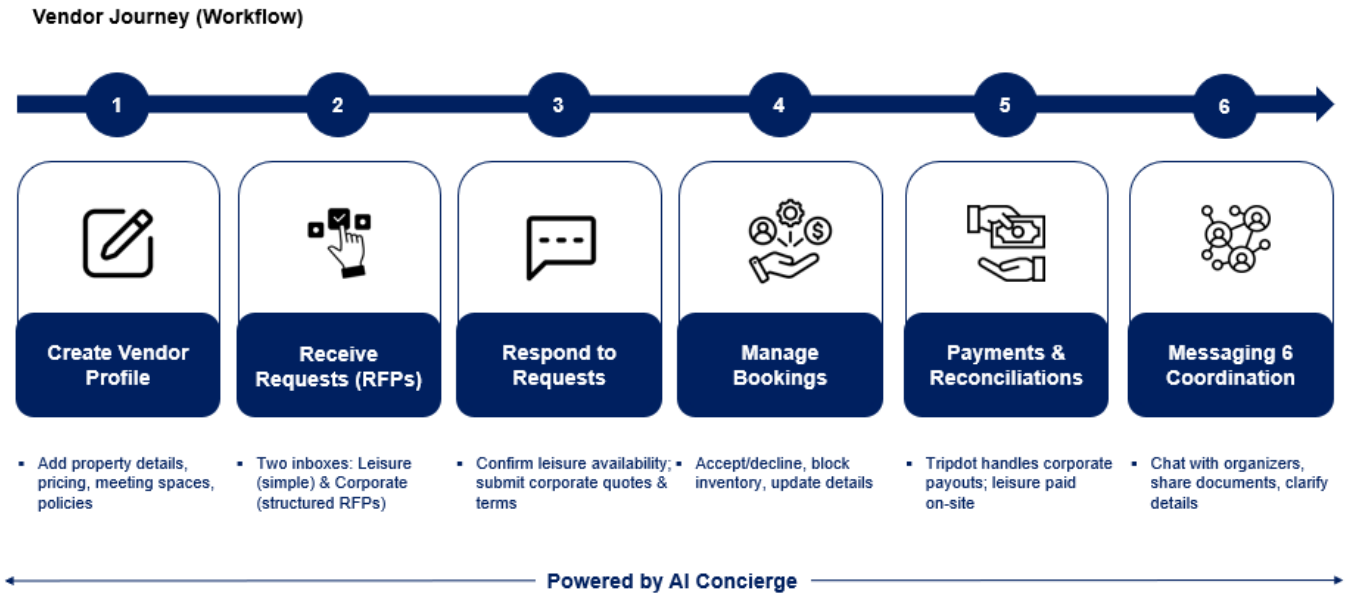
4.1. Leisure Group Travel Planning & Booking



4.2. Offsite Planning & Booking

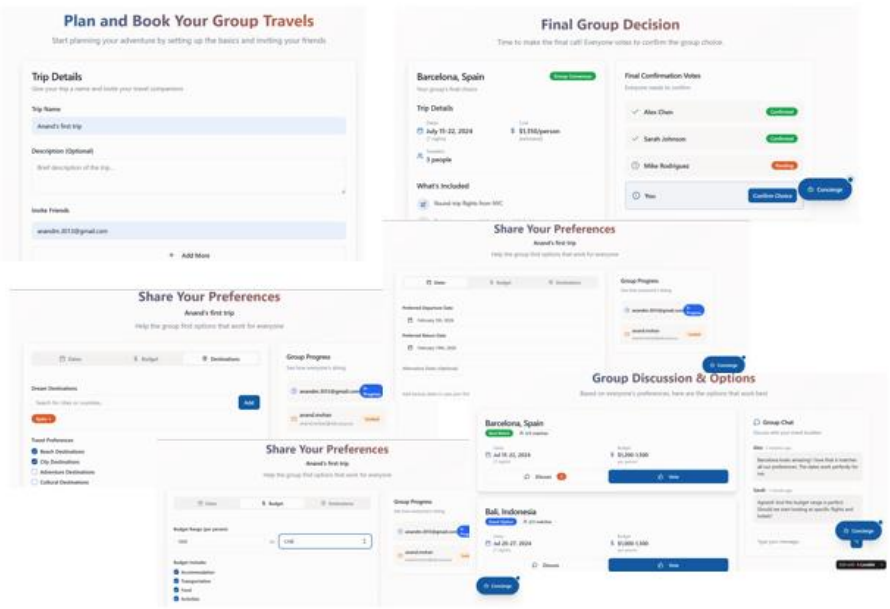


4.3. Vendor Management



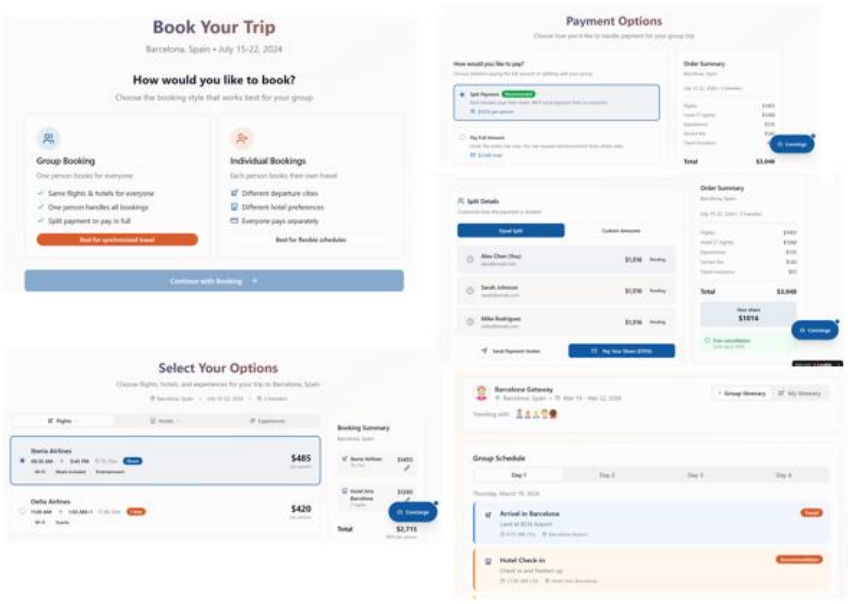
5. Product Screenshots:

5.1. Leisure Group Planning & Booking:



Planning & Coordination

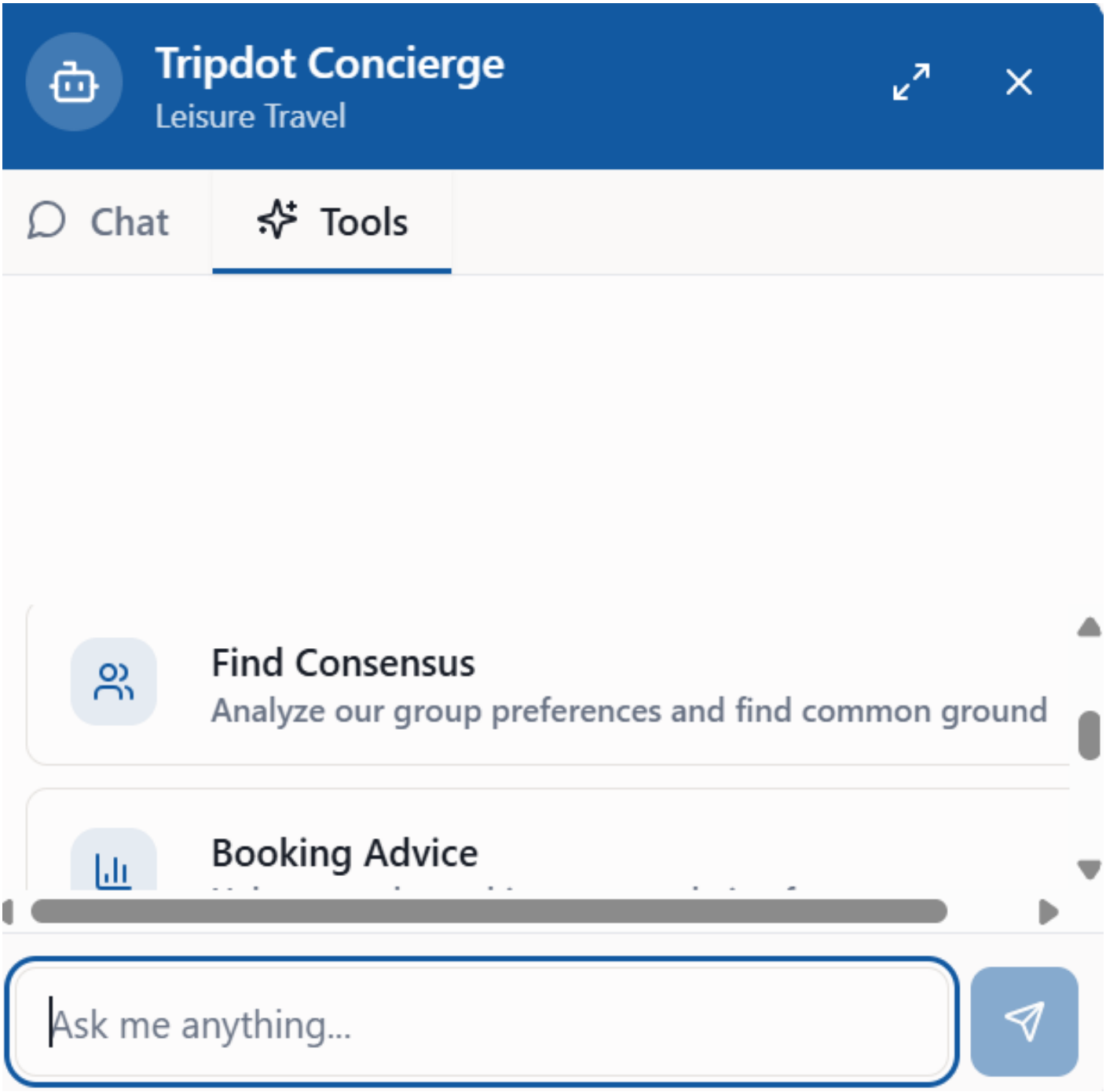
"As an organizer, I want to create a group of people travelling together, share preferences, vote, and reach a consensus in one platform."



Payment & Itinerary

"As a traveller, I want flexible payment splitting to avoid back-and-forth later, and clear visibility into both my personal and the group itinerary"

5.1.1. AI Assistant for Leisure Group Travel Planning:



AI Consensus Found

Your group has reached agreement!

Recommended: Hotel Arts Barcelona

Based on your group's priorities, Hotel Arts offers the best mix of beach access, nightlife proximity, and fits within everyone's budget at €93/person for 3 nights.

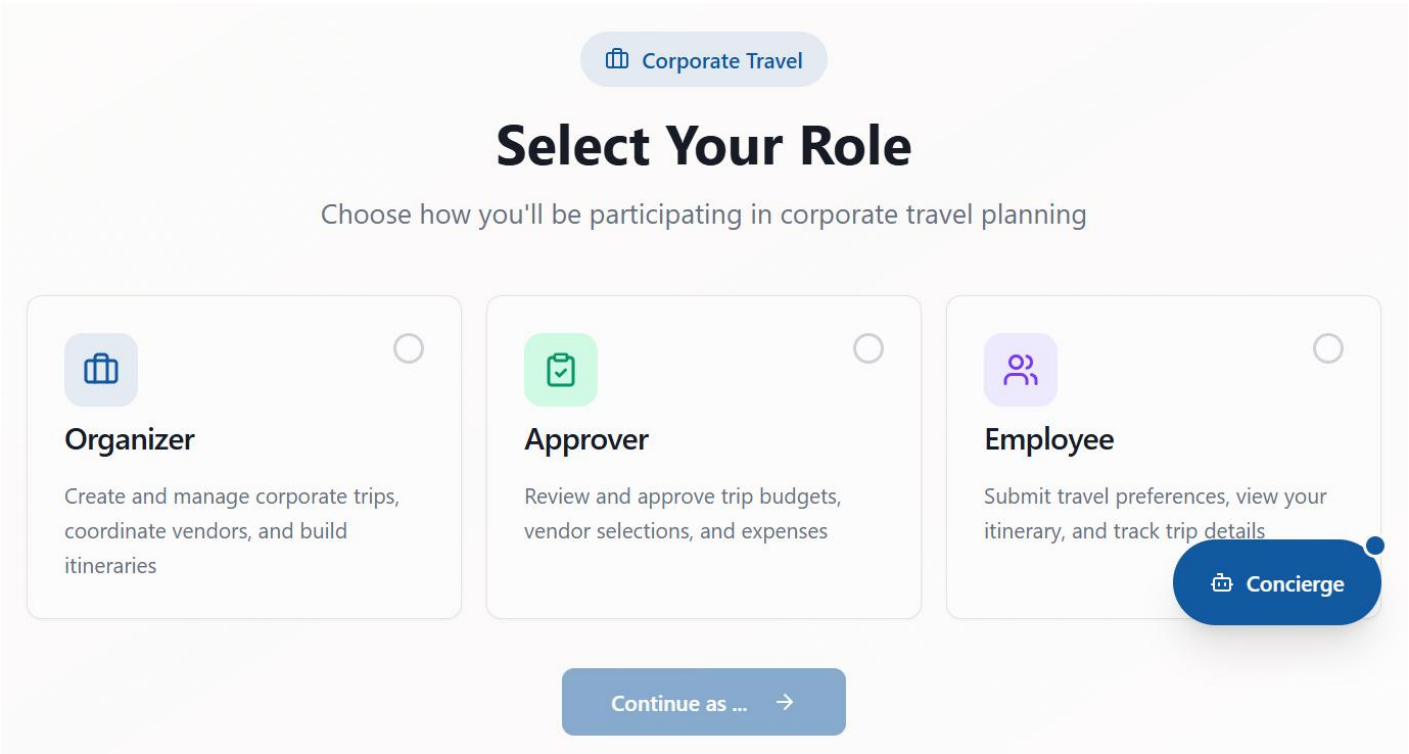
Dates: March 19-22 works for 5 of 6 members

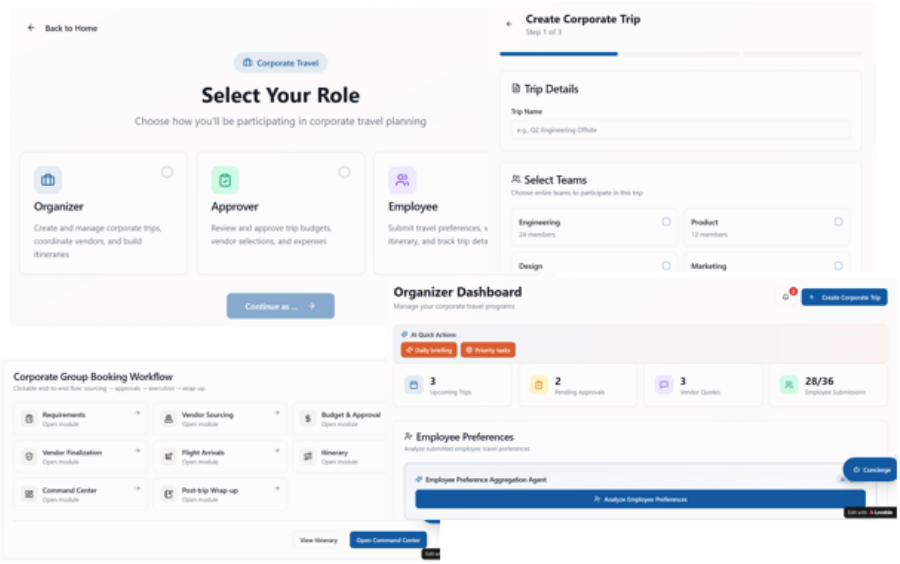
Budget: Average target is €580, Hotel Arts comes in at €558 total

Activities: Food experiences are the top priority, followed by beach time

All 6 members approve this plan!

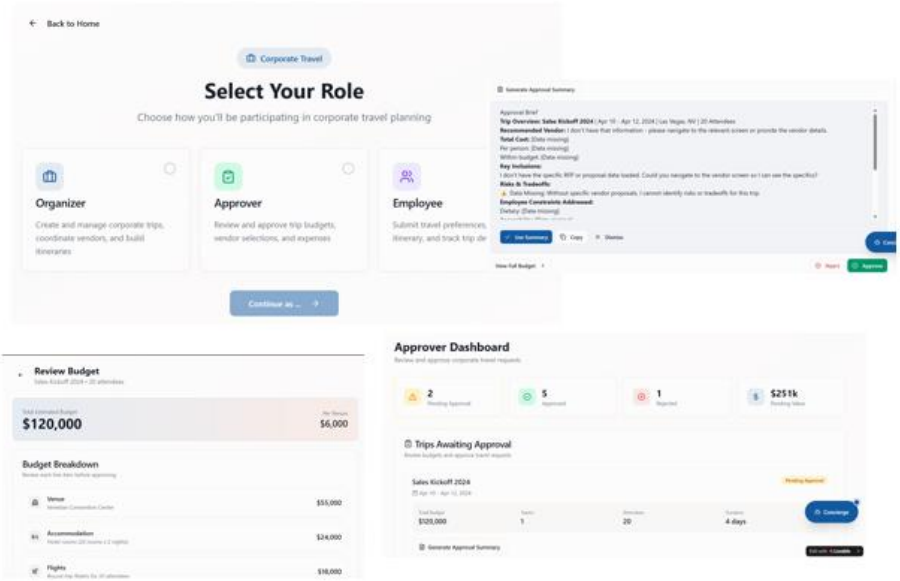
5.2. B2B (Offsite Planning & Booking):





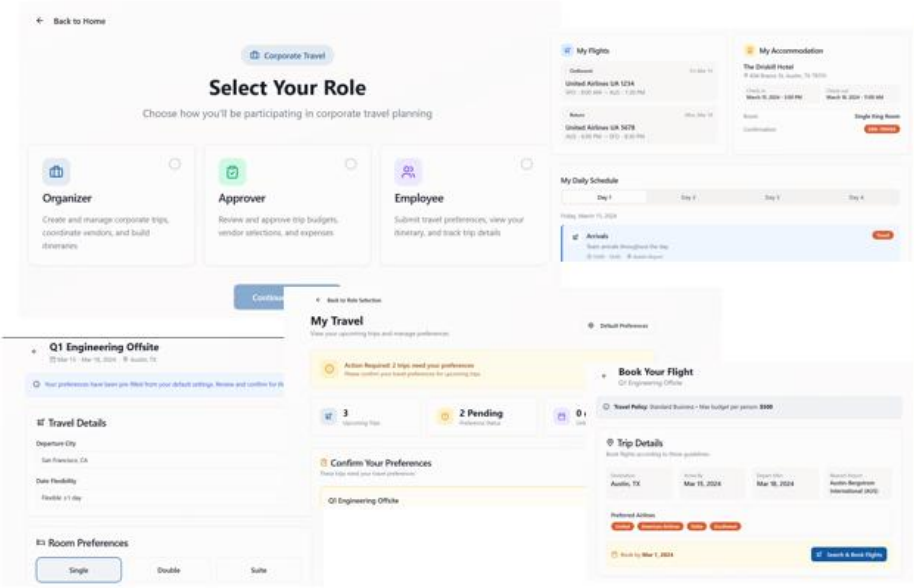
Organizer

"As an offsite organizer, I want centralized workflow for trip creation, vendor management, employee preferences consolidation, itinerary creation and visibility."



Approver

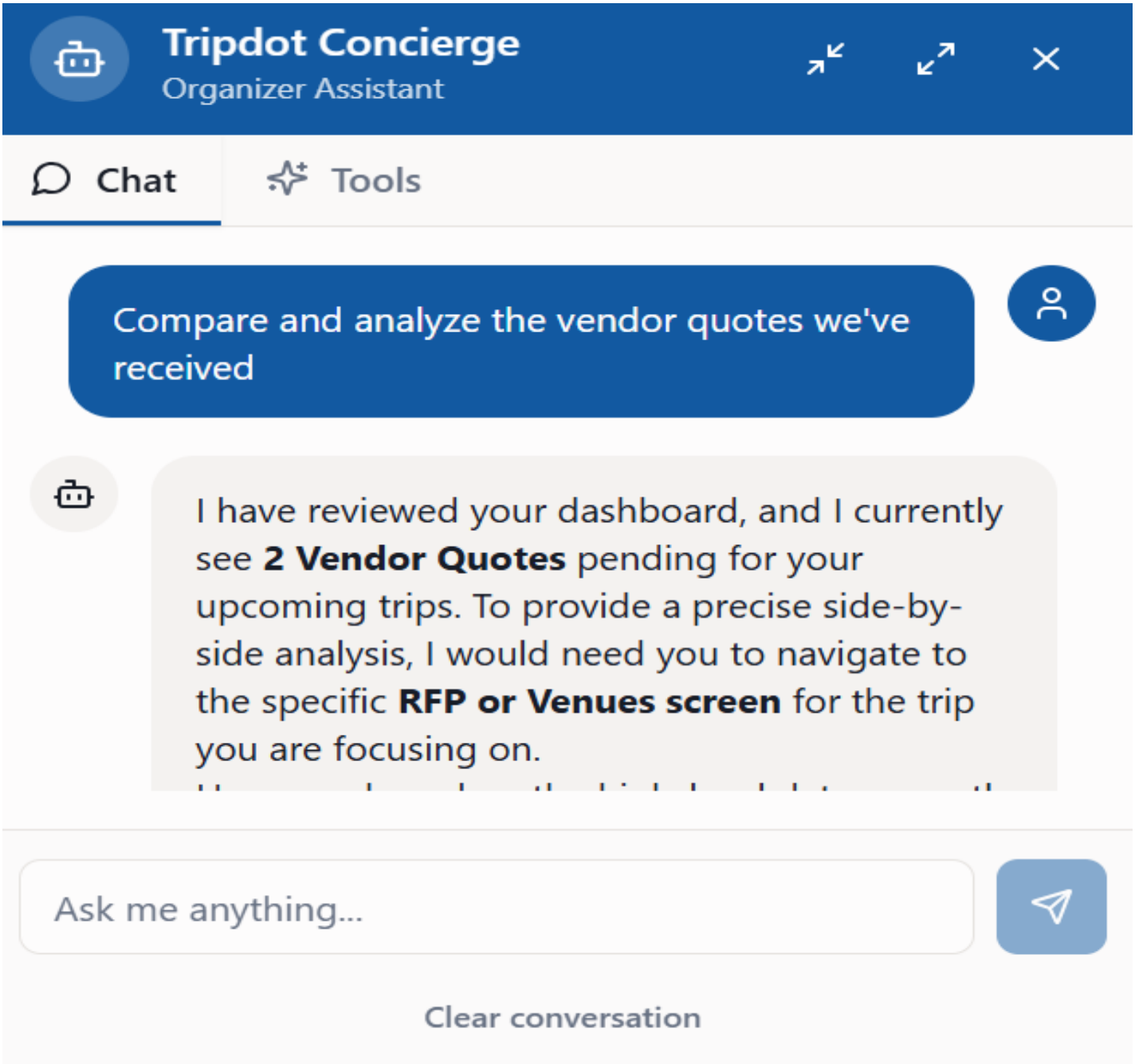
"As an approver, I want to see real time budget, structured approval, and consolidated costs"

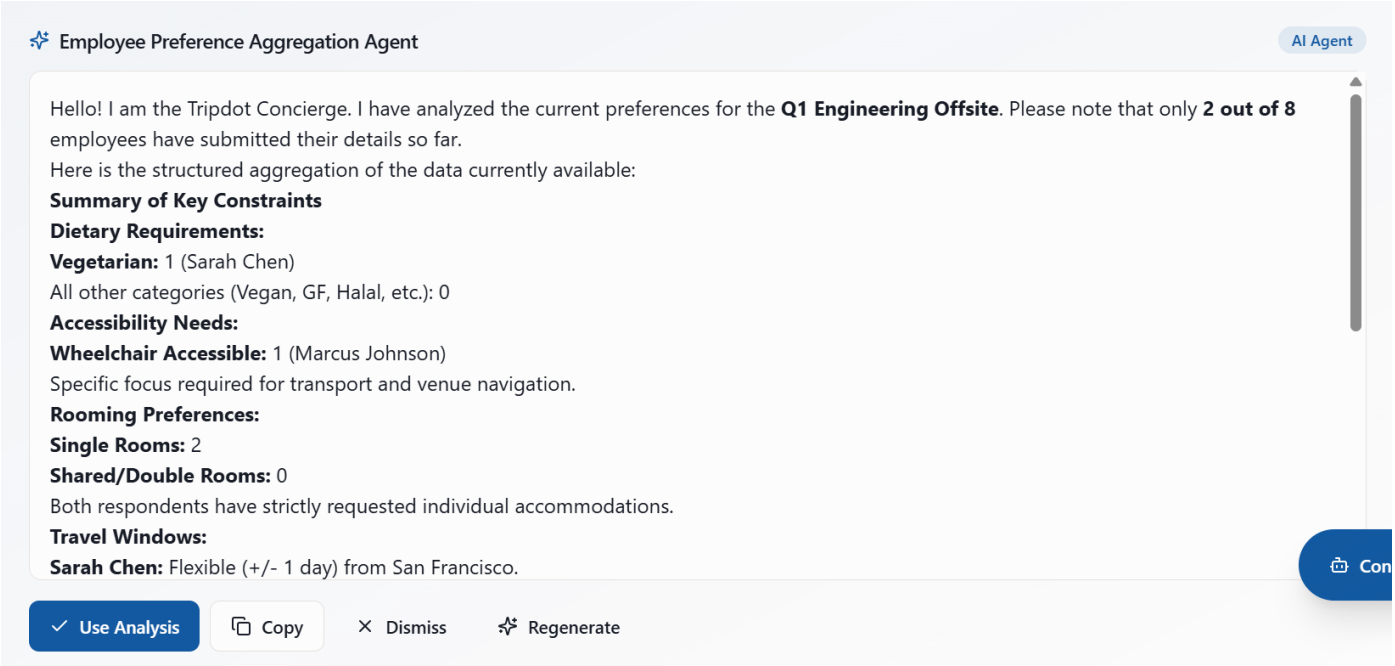


Employee

"As an employee, I want a central system to submit preferences and access personalized and real time itinerary."

5.2.1. AI Assistant for B2B Offsite Planning:





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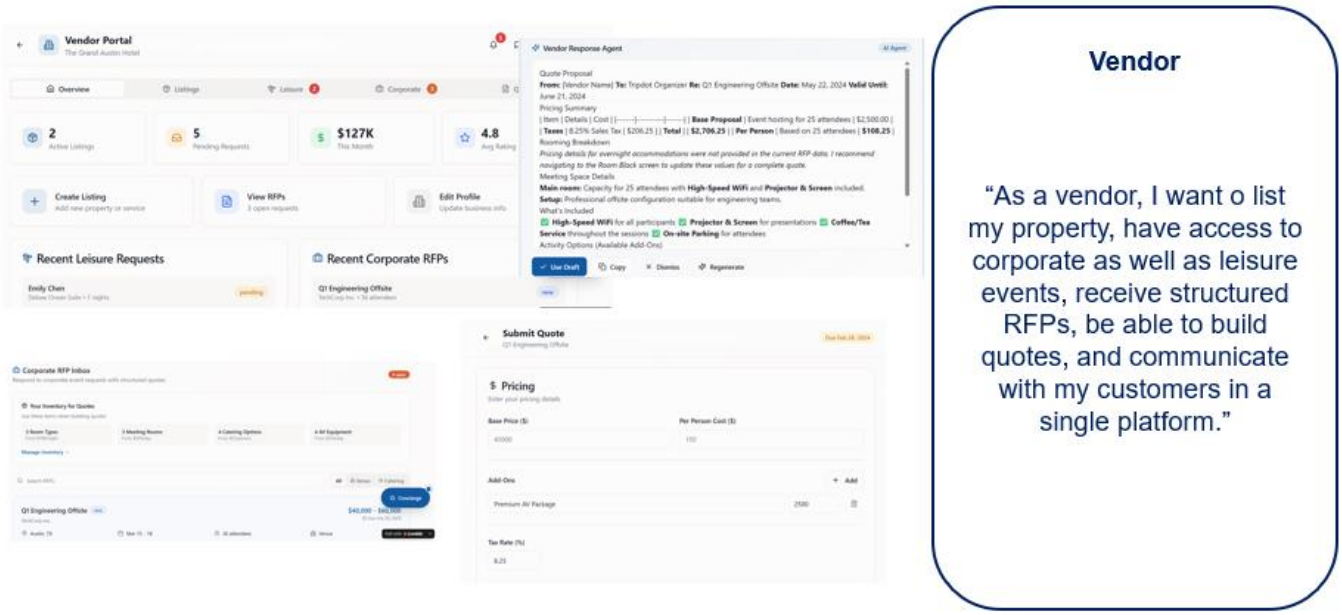

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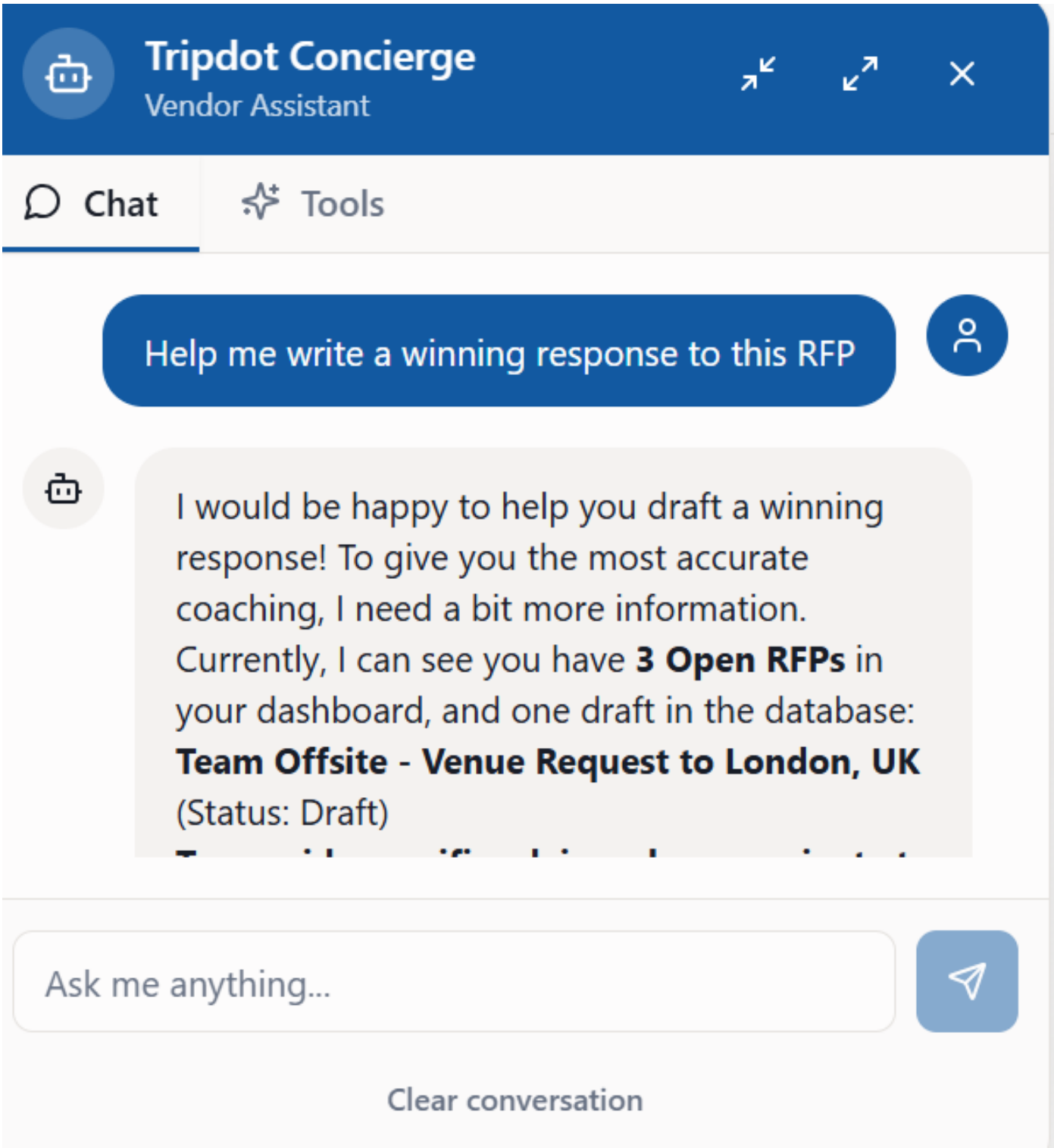

Vendor Response Agent

AI Agent

Quote Proposal

From: [Vendor Name] **To:** Tripdot Organizer **Re:** Q1 Engineering Offsite **Date:** May 22, 2024 **Valid Until:** June 21, 2024

Pricing Summary

| Item | Details | Cost | |------|---------|------| | **Base Package** | Accommodation & Meeting Space for 25 Attendees | $2,500.00 | | **Taxes** | 8.25% Sales Tax | $206.25 | | **Catering** | Coffee/Tea Service Included | $0.00 | | **Activities** | Professional On-site Amenities | Included | | **Total** | | **$2,706.25** | | **Per Person** | Based on 25 Attendees | **$108.25** |

Rooming Breakdown

Standard Occupancy: 25 attendees for the duration of March 15-18, 2024.

Specific room type pricing is available upon final selection of room blocks.

Meeting Space Details

Main Room: Equipped for 25 attendees with **Projector & Screen** included.

Connectivity: **High-Speed WiFi** provided throughout the venue.

What's Included

✅ **High-Speed WiFi** for all engineering sessions ✅ **Projector & Screen** setup ✅ **Coffee/Tea Service**